

UNIT IV: Express & Angular

- **Q1: Explain Routing in Express.** How do you handle GET and POST requests?
- **Q2: What is Data Binding in Angular?** Explain One-way vs Two-way binding.
- **Q3 (Guaranteed): Explain Angular Directives.** Differentiate between *Built-in* (like `*ngFor`) and *Custom* directives.
- **Q4: What are Angular Services?** Why are they used?

* Data Binding

- Data binding connects HTML (View) & TypeScript (Code)
- It helps to display data.
- It also helps to respond to user actions.
- It updates UI automatically
 - when data changes.

Types of Data Binding

1. Interpolation - Shows data using `{{ }}`
2. Property Binding - Binds values using `[]`
3. Event Binding - Handles events using `()`
4. Two-way Binding - Binds both ways using `[]()`

⇒ It keeps UI & data in sync.

⇒ No need to ~~man~~ manually update DOM.

⇒ ~~It~~ It makes app dynamic

⇒ It makes app interactive.

* Built-in Directives

- Directives are special keywords in Angular.
- It change how elements appear or behave.
- Angular provides built-in directives.
- Directives are special instructions in Angular.
- It can change the appearance of elements
- It can change the behaviour of elements.

⇒ There are 3 types of built-in directives:

- Structural Directives
- Attribute Directives

1. Structural Directives

- It is used to change structure
- add elements
- Remove elements
- Start with a * symbol.

Examples:

- *ngIf
- *ngFor
- *ngSwitch

2. Attribute Directives

- Used to change appearance or behaviour of an element.

examples:

- ngClass
- ngStyle
- required

* Custom Directives

- Custom Directives are user-defined directives.
- Custom Directives are created by developers.
- These are used when built-in directives are not enough.
- These are mostly attribute directives.
- They are created using the @Directive().
- You can create your own directive.

ng generate directive
directiveName

- It is used to apply styles.
- used to apply effects.
- Custom directives are useful when built-in directives are not enough.
- ⇒ Angular lets you create your own directives called Custom Directives.
- You can use custom directives to add styles, effects.
- Useful in large apps.